

# PAVAN KUMAR NAYAKANTI

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## PROFESSIONAL SUMMARY

Engineering leader with 20+ years heading cloud-native infrastructure automation and data platform operations across global capital markets institutions (Charles Schwab, RBC, BMO, ADP). Proven at leading and scaling DevOps and automation engineering teams across distributed geographies -- from 4 to 40 engineers -- while owning the full cloud operations lifecycle: provisioning, IaC governance, SLO management, automated remediation, and compliance in regulated environments. Deep expertise in Azure and GCP cloud-native operations, Infrastructure as Code (Terraform, Ansible), and observability platforms (Prometheus, Grafana, Dynatrace). Track record of translating operational challenges into scalable automation solutions that deliver measurable reliability improvements. Bringing capital markets domain depth to Deutsche Borse Group's Big Data & Advanced Analytics infrastructure transformation.

## CORE COMPETENCIES

Cloud-Native Data Platform Operations (Azure - GCP)

Infrastructure as Code (Terraform - Ansible)

Big Data Infrastructure Automation

DevOps Engineering Team Leadership

SLO / SLA Management & Automated Remediation

Compliant Cloud Operations (SOX - GDPR)

Capital Markets Domain Expertise

Distributed Team Leadership (India - Canada - US)

## WORK EXPERIENCE

IRIS Software Sep 2020 -- Present

Sr. Specialist -- Cloud Practice & SRE Lead | India | Clients: ADP, Charles Schwab, BMO

- Owned **cloud-native operations lifecycle** across multi-client platforms -- provisioning, IaC governance, scaling, IAM, monitoring, and automated remediation -- for enterprise banking workloads on AWS and Azure.
- Led **Infrastructure as Code** strategy using **Terraform** and **Ansible**, automating resource provisioning and configuration across ADP, Schwab, and BMO environments -- improving release velocity and operational resilience to 99.9%.
- Defined and governed **SLOs, SLIs, and error budgets** across multi-client cloud platforms, establishing automated remediation workflows and reliability thresholds aligned to business commitments.
- Deployed full-stack **observability** using **Prometheus**, **Grafana**, and Dynatrace across ADP's platform -- enabling real-time alerting, root cause analysis, and proactive incident prevention; **reduced MTTR by 40%+** via Python and Ansible automated remediation.
- Architected enterprise DevOps tooling including the Cloud Platform Maturity Index (CPMI) and DevOps as a Service -- defining measurable **engineering KPIs** and automation maturity benchmarks across three major banking clients.
- Led BMO's end-to-end AWS cloud migration -- delivering **AWS CDK**-enabled deployment pipelines across ECS Fargate and Lambda with High Availability and Disaster Recovery by design; drove FinOps practices reducing cloud waste across the AWS estate.
- Designed and deployed **GenAI-based data pipeline assistants** on AWS Bedrock for BMO's BCRG and LTS pipelines -- enabling risk summarisation and Q&A over structured and unstructured financial data.
- Scaled ADP's SRE organisation from 4 to 40 engineers** across India, Canada, and the US; simultaneously built Charles Schwab's 18-member Platform Engineering team from the ground up within a single quarter.
- GCP landing zone preparation and workload migration PoCs for Charles Schwab -- evaluating **GCP-native services** for cloud-native deployments and executing cross-platform artifact alignment across TD Ameritrade and Schwab post-merger.

## IRIS Software

Jun 2018 -- Sep 2020

### Automation Technology Specialist | Toronto, Canada | Client: Royal Bank of Canada Insurance

- Established SRE principles and **automated operational guardrails** within RBC's DevSecOps programme -- introducing observability standards, incident classification, and reliability-focused deployment gates across 250+ applications.
- Automated artifact retrieval, approval, and compliance workflows using **Python and Ansible** -- eliminating manual intervention and establishing audit-ready change evidence pipelines for SOX/GDPR regulated environments.
- Inducted and operationalised the **DORA framework** across RBC Wealth Management and Insurance -- establishing Deployment Frequency, Lead Time for Changes, Change Failure Rate, and MTTR as primary engineering health KPIs.
- Governed enterprise-wide cloud operations and change management across 250+ applications -- standardising deployment workflows, enforcing compliance, and ensuring audit-ready change transparency.
- Integrated **SAST, DAST, and IAST** scanning (SonarQube, JFrog Artifactory) into CI/CD pipelines, enforcing shift-left security and dependency management at scale.
- Led enterprise migration of 250+ applications across 60+ teams from legacy SCM to GitHub -- enabling modern CI/CD and self-service provisioning. **Reduced application onboarding from 5 weeks to 5 hours (98% reduction).**
- Implemented Splunk-based log aggregation integrated with CI/CD workflows, enabling proactive detection and rapid incident response across production environments.

## IRIS Software

Feb 2017 -- Jun 2018

### Automation Architect | Toronto, Canada | Clients: Insurance & Capital Markets

- Designed fault-tolerant, container-orchestrated deployment pipelines using **Kubernetes and Helm** -- enabling zero-downtime rollouts across capital markets platforms.
- Standardised **Python-based automation frameworks** for enterprise workloads and scaled an offshore delivery team across regulated insurance and capital markets domains.
- Led enterprise automation across 80+ applications and 250+ components, pioneering Cognitive and Intelligent Process Automation (CPA/IPA) frameworks.

## TechMahindra Pvt. Ltd.

Jul 2015 -- Feb 2017

### Technology Manager | Noida, India | Client: Adobe

- Led end-to-end **data platform migration** from Informatica to SQL Server 2012 for Adobe -- schema redesign, ETL re-engineering, and enterprise-scale data validation across all production workloads.
- Executed zero-downtime data centre relocation for Adobe's business applications, owning the full migration plan, execution, and cutover across all production workloads.

## Microsoft Corporation

Jul 2007 -- Dec 2014

### Sr. SDET -- QA, Automation & Production Engineering | India

- Held dual mandate as Sr. SDET and Production Engineering owner -- accountable for full quality and operational lifecycle of Sales & Marketing platforms serving users across US, EMEA, and APAC; maintained **99.99% uptime over 7 years**.
- Built QA automation frameworks integrated with CI/CD pipelines; owned all non-functional objectives -- performance, availability, scalability, security -- across infrastructure and application tiers.
- Supported 200% user growth with no platform reliability degradation through cloud migration and self-healing architectures.

KEY DELIVERY HIGHLIGHTS

Cloud Platform Maturity Index (CPMI) & DevOps as a Service ADP / Schwab / BMO

Architected and delivered an enterprise cloud maturity framework and DevOps-as-a-service platform establishing measurable engineering KPIs -- automation maturity, deployment frequency, reliability indicators -- across three major banking clients simultaneously. Included Terraform-driven IaC, Ansible configuration automation, and Prometheus/Grafana observability stacks for 200+ resource types.

Terraform - Ansible - Prometheus - Grafana - Dynatrace - AWS - Azure - Python

BMO GenAI Data Pipeline -- AWS Bedrock BMO / IRIS

Designed and deployed GenAI-based assistants on AWS Bedrock for BMO's BCRG and LTS pipelines, enabling risk summarisation and interactive Q&A over financial data. Responsible for the infrastructure automation, IAM, and cloud operations layer supporting the data pipeline -- provisioning, scaling, monitoring, and compliance automation in AWS.

AWS Bedrock - ECS Fargate - Lambda - AWS CDK - IAM - CloudWatch - Python

modeas -- High-Throughput Cloud Deployment Platform ADP / IRIS

Delivered an enterprise deployment and operations platform handling 200+ builds/deployments per hour, eliminating 10,000+ manual operational hours annually and accelerating delivery cadence by 600%. Built on Kubernetes/Helm with full observability (Prometheus, Grafana) and self-healing automation -- directly analogous to the infrastructure automation layer Deutsche Borse's BDAA team requires.

Kubernetes - Helm - Terraform - Ansible - Prometheus - Grafana - Python - Jenkins

RBC DevSecOps Programme -- 250+ Applications at Scale RBC / IRIS

Led enterprise-wide automation and compliance enforcement across 250+ application portfolios for Royal Bank of Canada -- integrating SAST/DAST, automated compliance guardrails, DORA metrics, and self-service CI/CD pipelines. Reduced application onboarding time from 5 weeks to 5 hours through the Robo Deployment Framework.

Jenkins - GitHub - SonarQube - JFrog Artifactory - Python - Ansible - Azure DevOps - Splunk

EDUCATION

|                                    |                     |      |
|------------------------------------|---------------------|------|
| Masters in Artificial Intelligence | Woolf University    | 2024 |
| General Management Program (GMP)   | ISB Hyderabad       | 2022 |
| M.Sc in Computer Science           | Osmania University  | 2002 |
| B.Sc in Electronics                | Kakatiya University | 2000 |

CERTIFICATIONS

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|------------------------------------------------------------------------------------------------------|--------|
| Google Cloud Certified -- Professional Cloud Architect (PCA)                                         | 2026   |
| Microsoft Certified: Azure Solutions Architect Expert (AZ-305)                                       | Active |
| AWS Certified Solutions Architect -- Associate (SAA-C01)                                             | 2026   |
| Certified Kubernetes Application Developer (CKAD)                                                    | 2025   |
| Microsoft Certified: Azure AI Engineer Associate (AI-102)    Azure Data Scientist Associate (DP-100) | Active |
| Certified SRE (iSRE)    ITIL(R) IT Service Operations    PMI Agile Certified Professional (PMI-ACP)  | Active |

TECHNICAL SKILLS

**Cloud Platforms:** Microsoft Azure (AKS, Azure DevOps, Data Factory), GCP (GKE, BigQuery, Dataproc, GCS), AWS (ECS, Lambda, Bedrock, CDK)

**IaC & Automation:** Terraform, Ansible, AWS CDK, Helm, Python, Bash, PowerShell, Groovy

**Container & Orchestration:** Kubernetes, Docker, Helm, ECS Fargate

**Observability:** Prometheus, Grafana, Dynatrace, Splunk, AppDynamics, MS Application Insights, Telegraf

**Data Platforms:** Databricks Fundamentals - in progress , Apache Spark, Delta Lake, SAP HANA, SQL Server, Oracle, SSIS, ETL pipelines

**CI/CD & DevSecOps:** Jenkins, GitLab, GitHub Actions, SAST/DAST/IAST, SonarQube, JFrog Artifactory, SOX/GDPR compliance

**Reliability & ITSM:** SLI/SLO/Error Budgets, DORA Metrics, ITIL, ServiceNow, Chaos Engineering (Keptn), Incident & Change Management